



Documentation Quality Compliance Checker

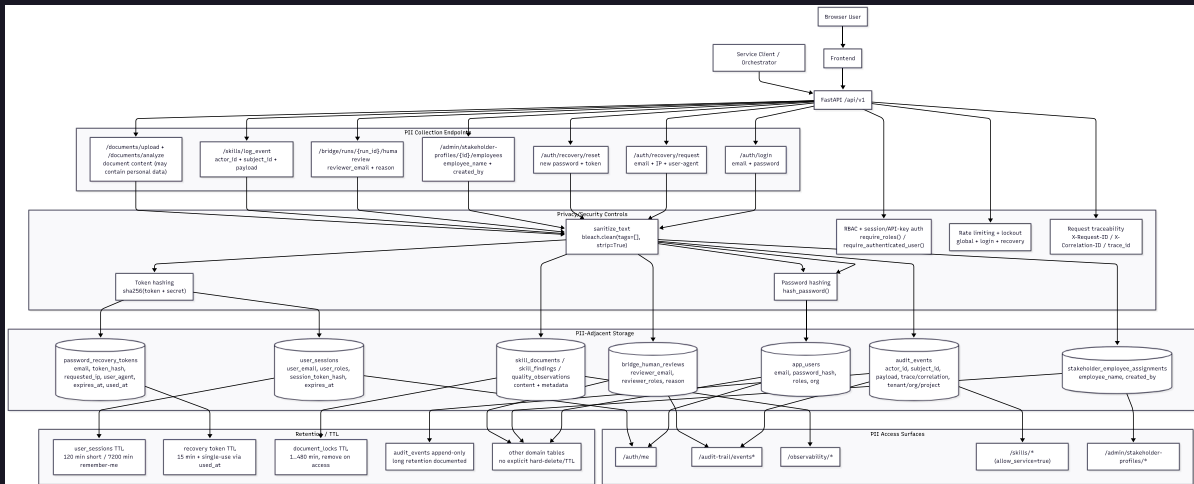
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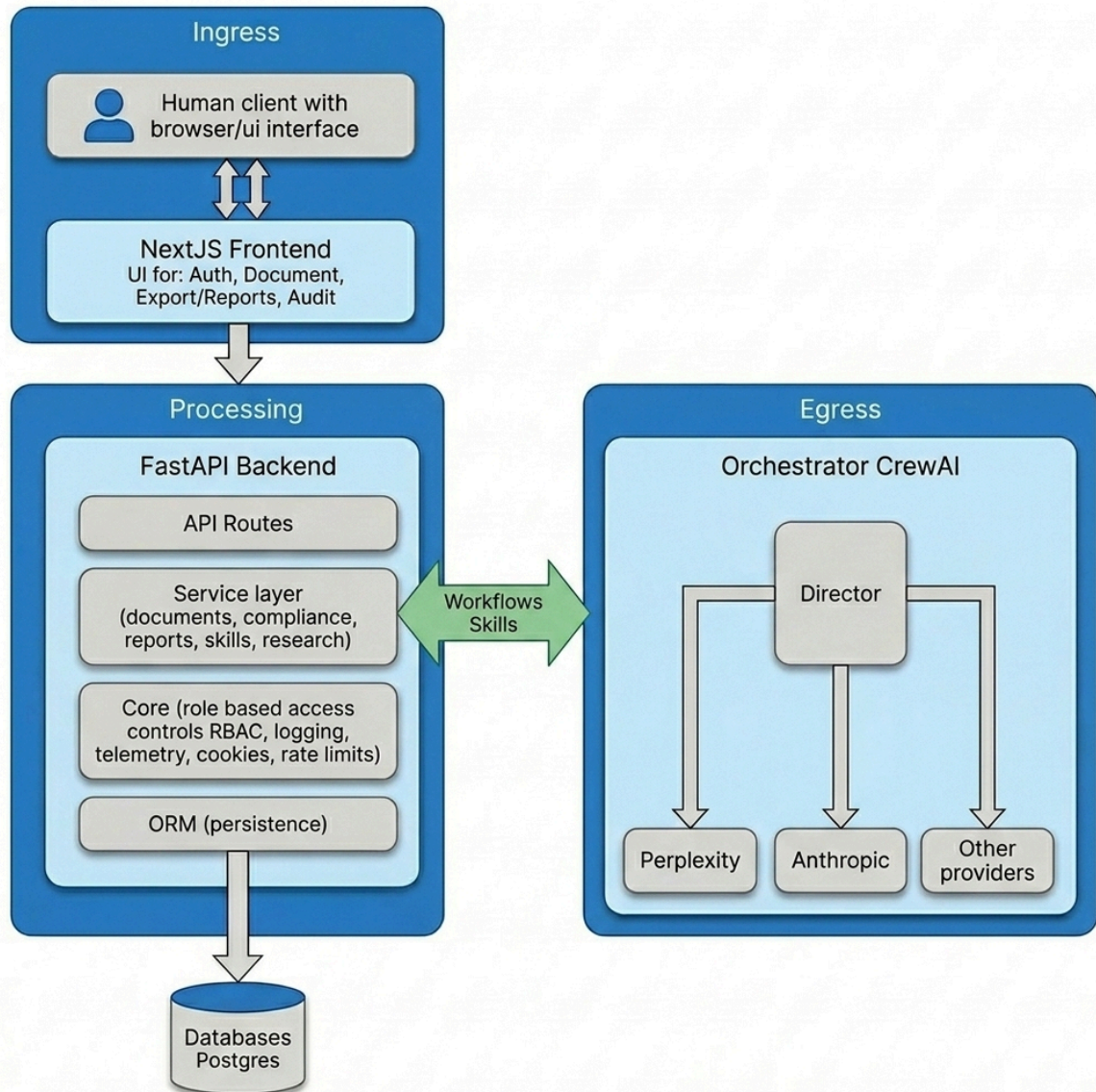
Project & use case

- What we audited
- Problem solved
- Architecture at a glance
- Why privacy matters

- **Scope:** Documentation quality compliance checker performs AI-assisted QM audits of technical docs and governance docs (SOPs)

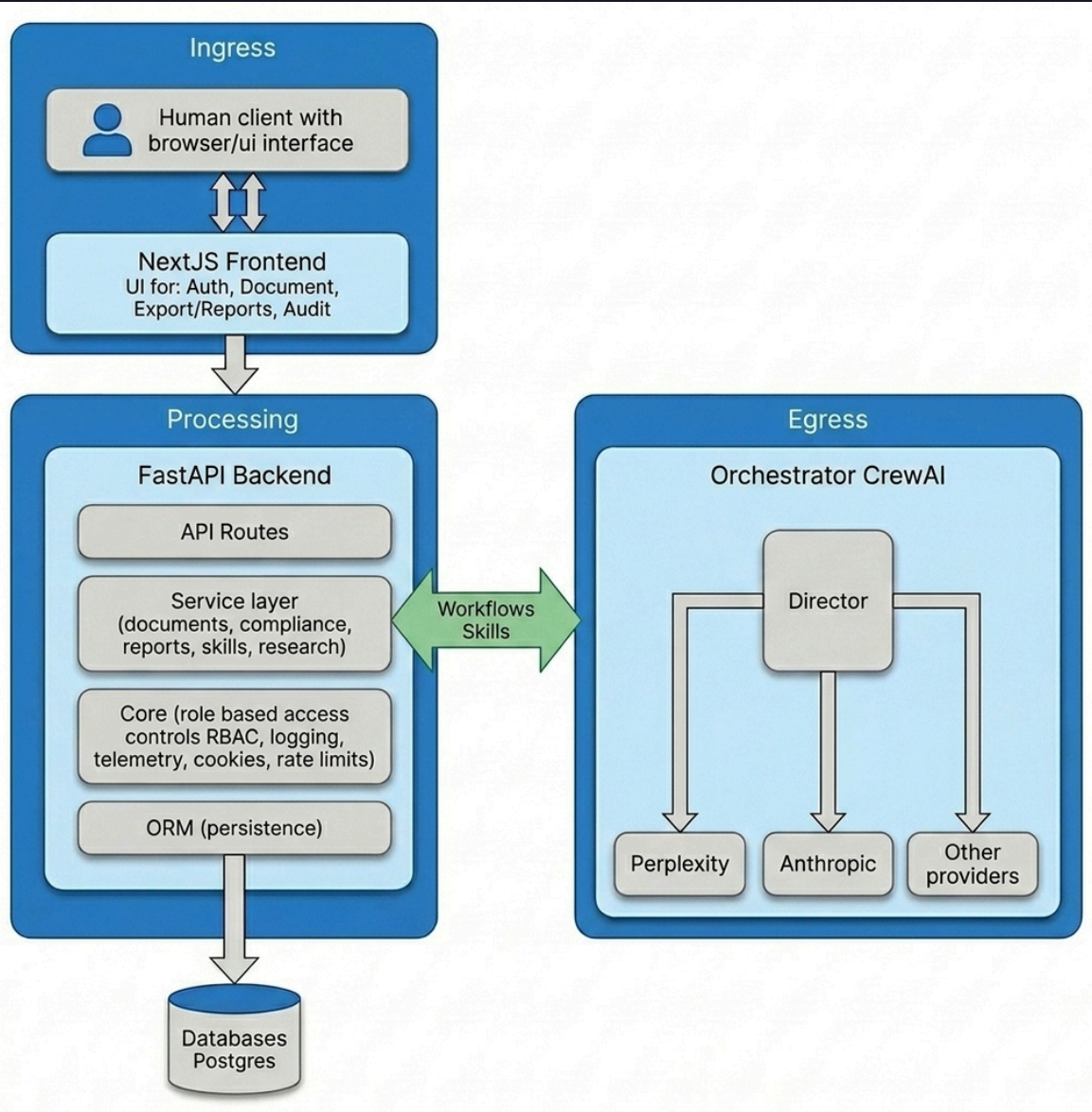
- **Audit:**
<https://willingc.github.io/compliance-checker/>





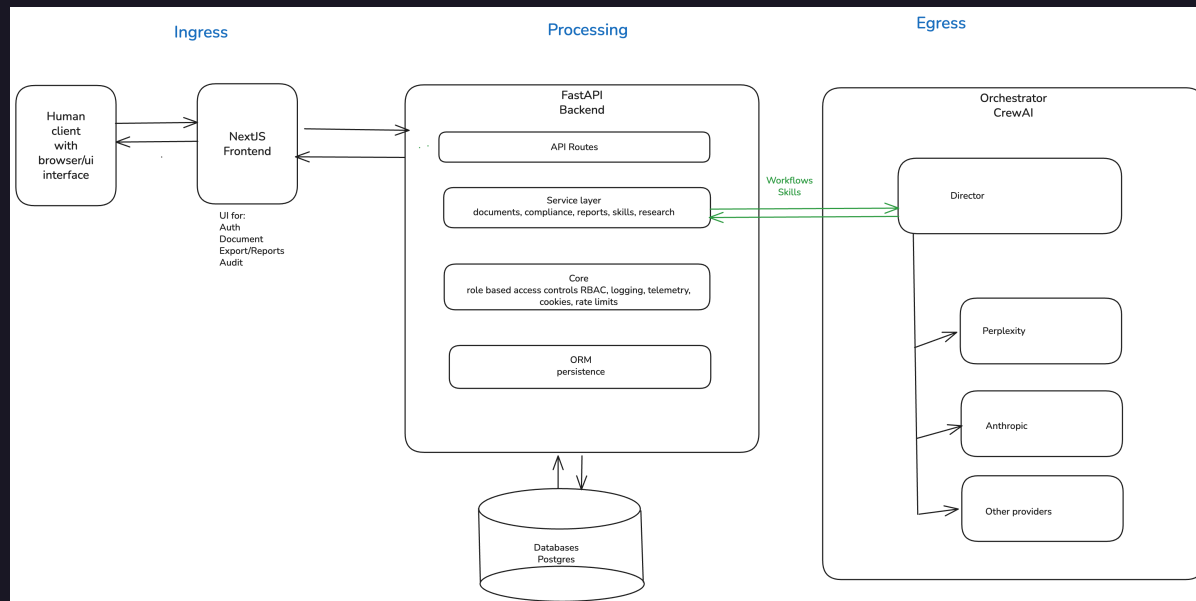
Problem solved

- Teams must prove documentation meets quality and regulatory expectations
- Manual QM review is slow, inconsistent, and hard to scale
- AI agents can help — but introduce new privacy and data-flow risks



High-level architecture

- Human operator
- Next.js frontend
- FastAPI backend
- PostgreSQL database
- Orchestrator and agents



Privacy matters

- User accounts, roles, and session data (GDPR-relevant)
- Document content and audit trails may contain sensitive business data
- Egress to LLM providers and telemetry expand the trust boundary

Top privacy risks

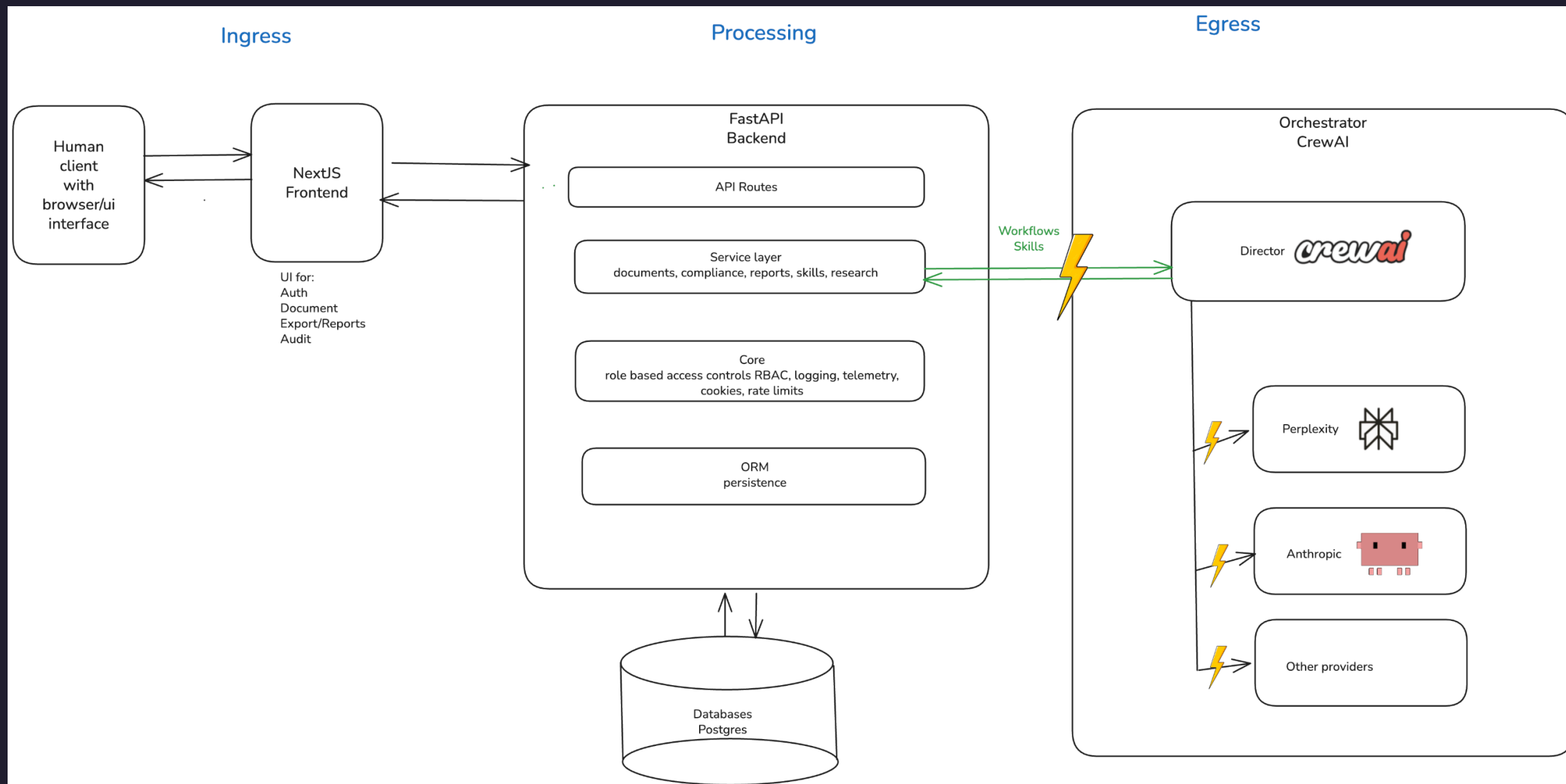
1. Egress traffic flow data to director and external models
2. Application telemetry and workflow tracing
3. Role-based access controls and GDPR compliance

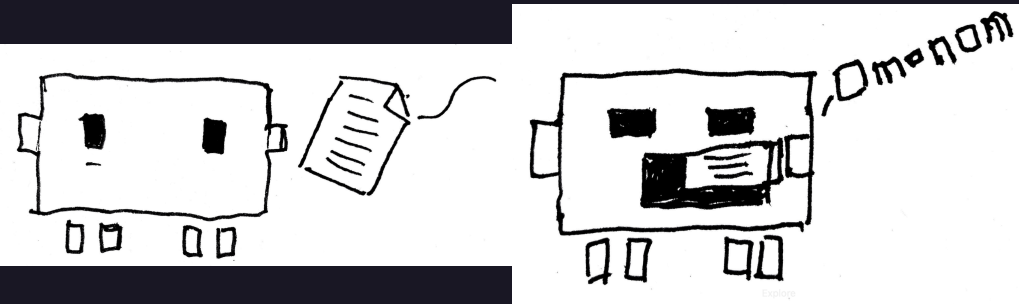
Up next: In-depth risk analysis

Let's look at each privacy risk...

- its sensitive data
- why the risk matters
- the priority of the risk
- **mitigations** of the risk

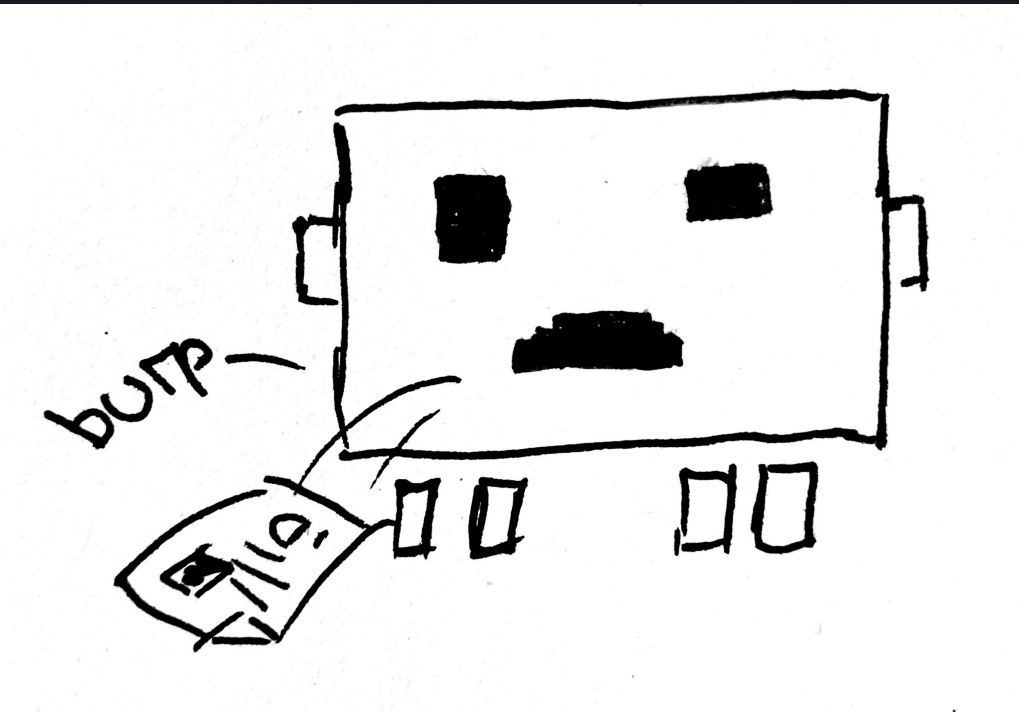
Egress exposes privacy data



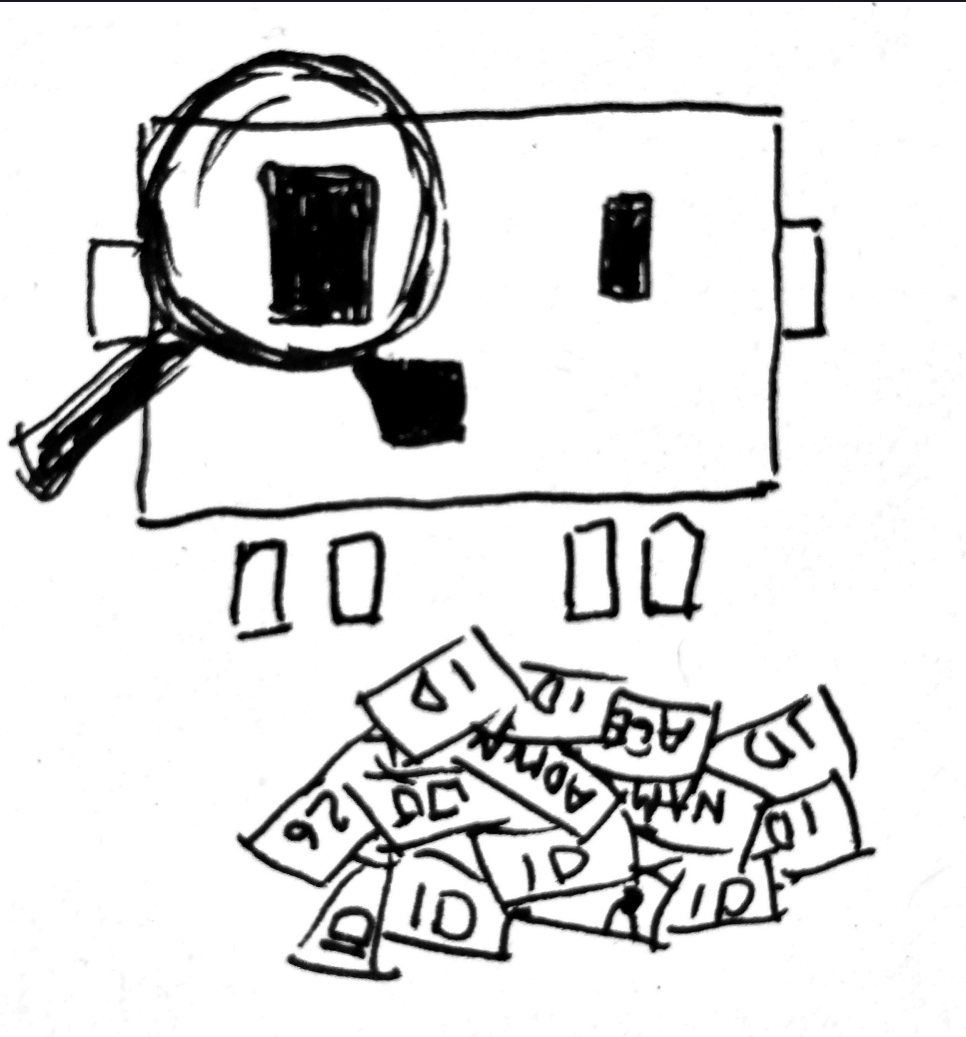


Sensitive data

- External model: transfer of potentially personal prompt/output data
 - **Data:**
 - Names, emails, stakeholder assignments, reviewer identifiers, document passages copied into prompts, generated summaries that may restate personal data
 - **Why Sensitive:**
 - Leaves the primary application context during model inference (current external-provider path)



- Over-collection and -retention in model observability traces
 - **Data:**
 - provider, model_used, trace_id, correlation_id, latency/tokens, rich payload entries, prompt/output snapshots
 - **Why Sensitive:**
 - Enables reconstruction of *who* triggered *what* model action and *when*
 - can become re-identifiable when combined with audit tables

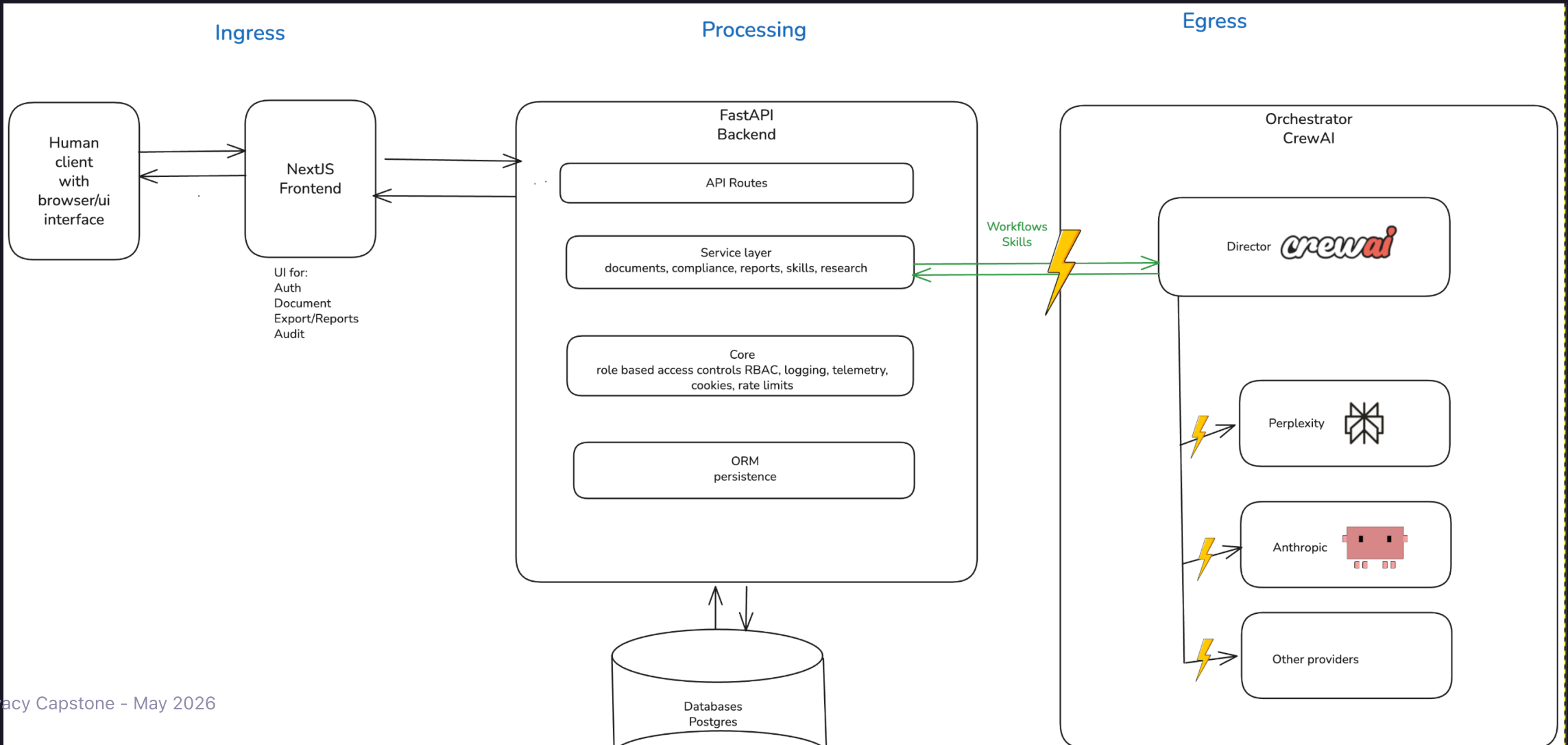


- Model credentials and routing configuration
 - **Data:**
 - API keys, adapter routing flags, provider selection settings
 - **Why Sensitive:**
 - Compromise enables data exfiltration or unauthorized model usage

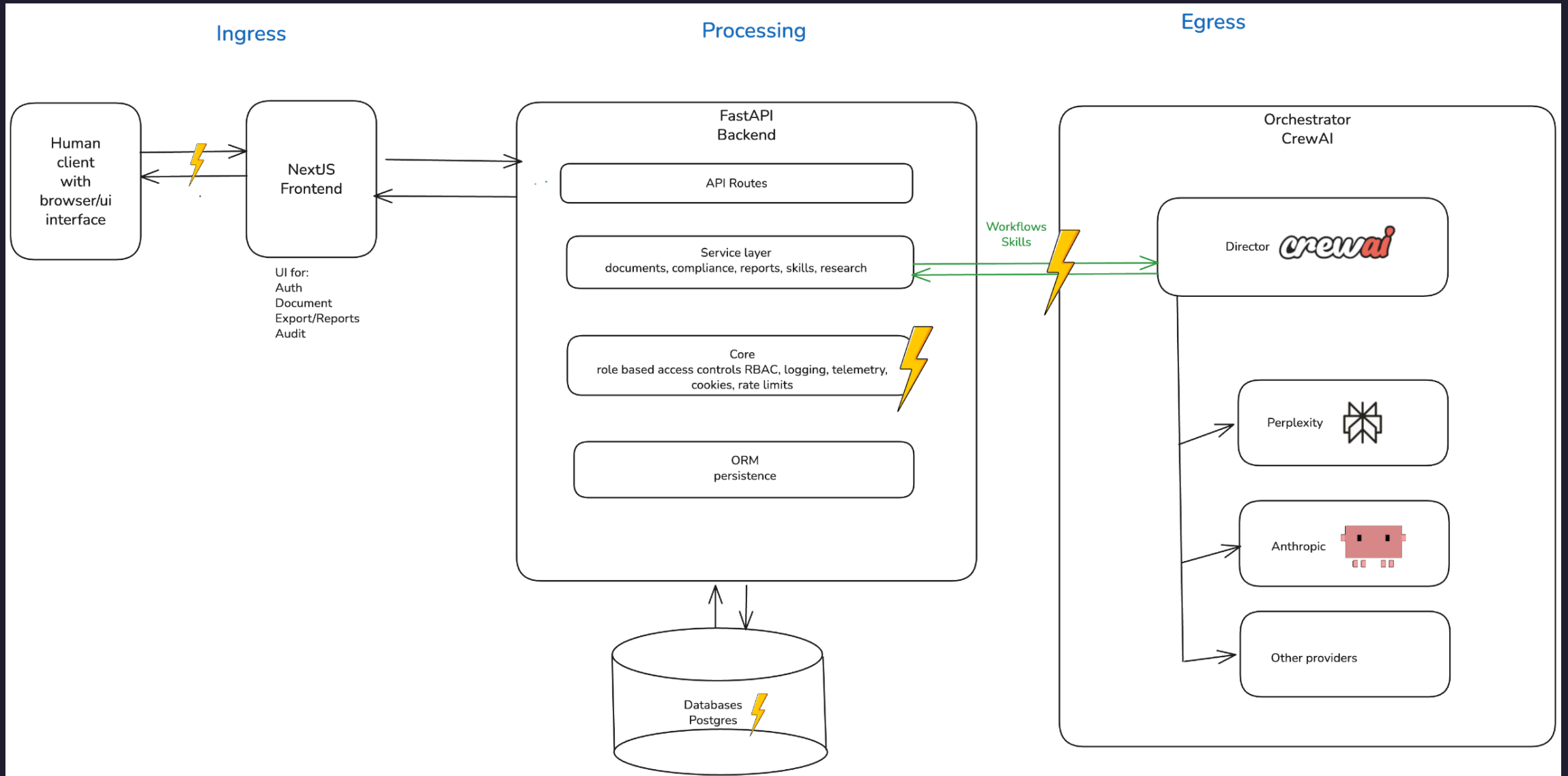
Mitigation

Egress traffic flow data to orchestrator and models

TODO: add happy mitigated image here



Audit logs/reports and telemetry expose privacy data



Sensitive data

- Raw LLM prompt and output in *quality_observations.payload*
 - **Data:**
 - *payload.provider, payload.model_used, payload.llm_temperature*
 - *payload.llm_prompt* (document text submitted by user, stakeholder names, reviewer assignments)
 - *payload.llm_output* (generated compliance summaries restating personal context)

- ○ **Why Sensitive:**
 - Prompt context is assembled from user-submitted documents which routinely contain names, emails, project identifiers, and business-sensitive content
 - The output may restate or summarise that content
 - Both are persisted to a queryable table

- Audit event payload in `audits_events.payload`
 - **Data:**
 - `payload.roles` (user role at login), `payload.remember_me` flag, event-specific context fields, any free-form data inserted by orchestrator or Skills API callers
 - **Why Sensitive:**
 - Append-only and intended for long retention
 - No technical control prevents a caller from writing personal data into the payload column
 - Combined with `actor_id` (user email) it forms a rich personal profile

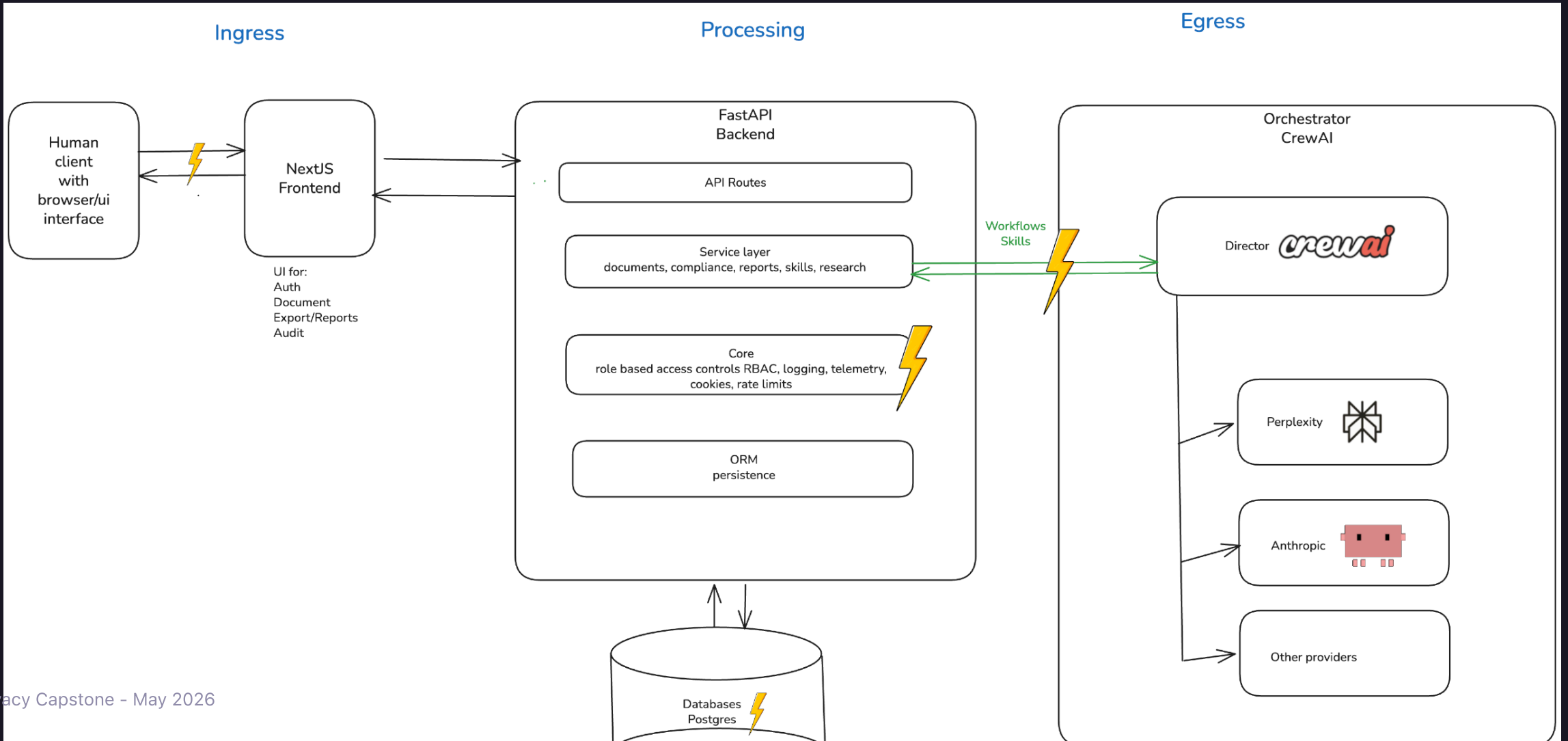
- OTEL (open telemetry) span attributes (exporter config)
 - **Data:**
 - HTTP path attribute (e.g., */api/v1/documents/doc-abc123* — document ID in path), user_agent, http.method, http.status_code, trace_id / span_id (linkable back to session)
 - **Why Sensitive:**
 - When TRACING_EXPORTER=otlp span data is sent to an external collector,
 - path values can embed document or session identifiers
 - user-agent can contribute to fingerprinting

- Frontend CSV export of prompt/output pairs
 - **Data:**
 - Exported observability *prompt_output_pairs.csv* file containing prompt, output, source_component, trace_id, timestamps
 - **Why Sensitive:**
 - Created on the user's local filesystem;
 - outside system access controls, retention policy, and audit log;
 - may contain personal data from documents processed during the export window

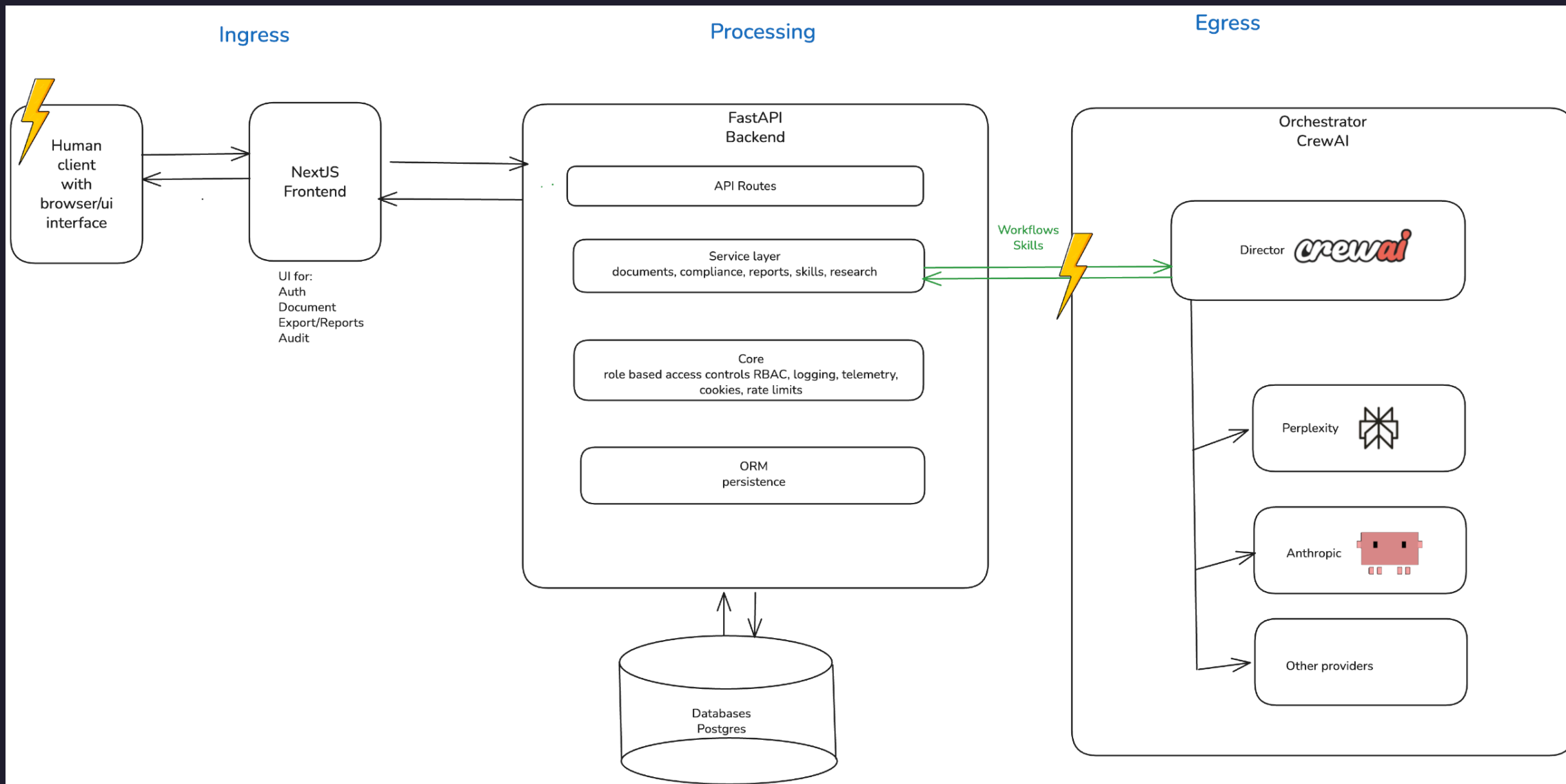
Mitigation

Application telemetry and workflow tracing

TODO: add happy mitigated image here



Misconfigured and stale controls give access to private data



Sensitive data

- User identity in session store
 - **Data:**
 - user_email (primary identifier), user_org, session session_id, expires_at, last_seen_at
 - **Why Sensitive:**
 - Directly identifies users and links their organisational role to access patterns and audit trails
 - Retained in PostgreSQL with no visible TTL-based purge policy

- Role assignments and permission scope
 - **Data:**
 - user_roles array per session row (e.g., ["qm_lead"], ["auditor"]), org isolation field user_org
 - **Why Sensitive:**
 - Reveals organisational responsibilities and access privileges
 - Can be used for social engineering or targeted attacks
 - GDPR data minimisation applies

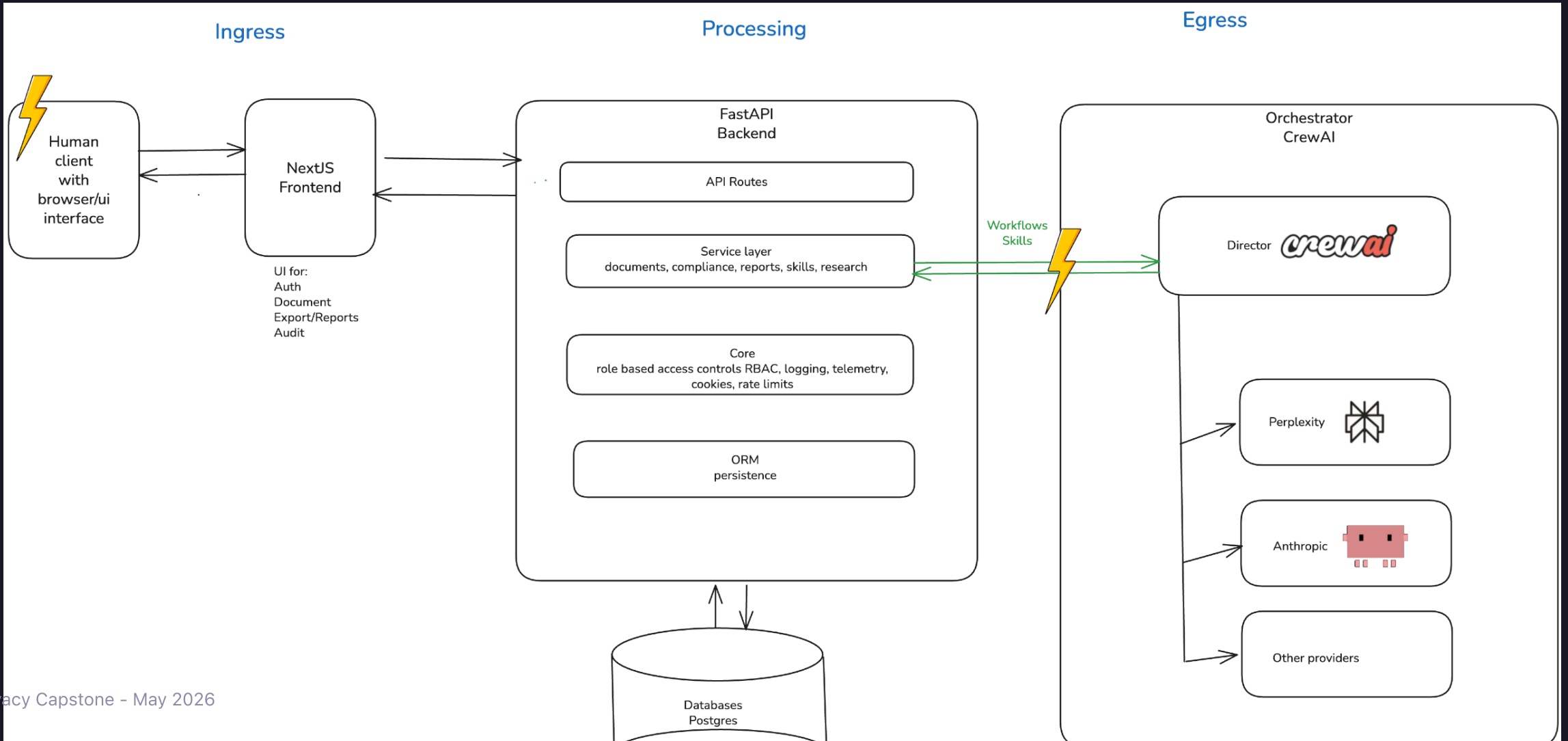
- Bootstrap/MVP credentials in environment configuration
 - **Data:**
 - AUTH_MVP_EMAIL, AUTH_MVP_PASSWORD, AUTH_MVP_ROLES, AUTH_MVP_ORG (env vars)
 - SECRET_KEY (API key secret)
 - **Why sensitive:**
 - Compromise of bootstrap credentials gives attacker a fully provisioned account with configurable roles
 - SECRET_KEY grants service-client access to skills and observability

- Access decision and audit context not separately persisted
 - **Data:**
 - Which role accessed which route at what time
 - 403 access denials
 - service-client route usage with payload summary
 - **Why sensitive:**
 - Absence of access-decision audit log prevents retrospective investigation of data-access incidents
 - Required for GDPR Art. 30 record of processing activities and breach response

Mitigation

Role-based access controls and GDPR compliance

TODO: add happy mitigated image here



Recommendations

- Top priority going forward
 - Why it matters
- Fit with company and risk strategy
 - Value created

Top priority going forward

- Meet the standard directives and too much to do manually
- Maintain tool for efficiency

Why it matters

- Insert why it is important (ROI, risk reduction, user impact, compliance, security)
- Too much for any one human to analyze 1000 page document
- Time for weeks and months for a document
- Changes if directives change
- Expensive specialists
- Basic insights, what has changed, where changed, policies affected

Fit with company and risk strategy

- GDPR / BSI in Germany regulations for software rules
- Main points are addressed for external audits

Value created

- Being audit

Questions & answers

- Audit scope and findings
- Risk prioritization and trade-offs
- Mitigations and open items

Thank you

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Privacy audit

<https://willingc.github.io/compliance-checker/>

Application

https://github.com/IloBe/doc_quality_compliance_check

